AI-Powered Agriculture Crop Advisory Assistant Architecture

Multi-Agent LangGraph Orchestration • Hybrid Retrieval • Classical ML • LLM-as-Judge

PHASE 1: PREPARATION (once, make setup)

saves cases, models, indexes to disk

Kaggle Crop/Soil/Weather CSVs

Merged ~19.6k cases (synthetic fallback)
Canonical schema mapping + cleaning

Data Processing & Feat. Eng

Tertile Cutoffs (thresholds.json)
Narrative Builder (CASE-XXXXXX)

Dual Storage & Vector Index

ChromaDB (cases__provider__model)
BM25 Okapi Index (Cache)

ML Model Training

Yield: GradientBoosting/XGBoost
Risk: Calibrated Classifier

PHASE 2: ANSWERING (every query, reads what Phase 1 saved)

React + Vite UI

Advisor, Compare, Eval, Feedback
Provider & Mode Selector

FastAPI Service

REST Endpoints (/api/v1/query)
CORS, Rate Limit, Lifespan

Input Guardrails

Length, SQL/Injection, Domain Check
Physical Bounds (pH, NPK, Weather)

Classification Agent

Intent, Crop, Region, Season
Regex / Deterministic Fallback

Hybrid Retrieval (RRF, k=60)

BM25 + Chroma, shared metadata filters
Prioritization: proximity, yield, recency, feedback boost

Parallel Analysis Agents

• Impact Analyst (What-If Sens.)
• Yield Model (Intervals)
• Risk Model (Suitability)

Recommender Agent

Synthesis of Candidates
Grounding in Retrieved Cases

Output Guardrails + LLM Judge

Case-ID & number verification
Pass/fail computed in code (no rewrite loop)

invoke_with_fallback: provider chain

1. Ollama (default)

llama3.1:8b chat
nomic-embed-text

2. OpenAI (fallback)

gpt-4o-mini chat
text-embedding-3-small

3. Degraded Mode (no LLM)

Regex classifier, template answer,
rule-based judge, BM25-only search

feedback boost

Feedback Loop (SQLite)

Ratings + helpful cases ->
bounded case/crop re-rank boost